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Stated or Sampled? Post-Training Keeps the Base Model's Differential Diagnosis but Removes Its Variation Between Calls

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Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We measure how post-training changes both readouts on DDXPlus, whose synthetic patients carry a rule-based reference differential, with base/post-trained pairs from three open families at 7–32B (Qwen2.5 and Qwen3, every released stage of OLMo 3 7B and OLMo 3.1 32B, and Mistral-7B), zero- and few-shot prompts, and real rare-disease cases. Post-training keeps the base model's view of each patient but removes most of its variation between calls: a post-trained list is closer to its base model's averaged list for the same patient than to that model's averages for other patients with the same diagnosis, and it barely changes from call to call. Sampled answers collapse (1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9), and with each model prompted at its best, post-training costs 13–30 points of pass@32; for one of two models tested, zero-shot sampling at a higher temperature recovers part of this. One stated list improves robustly only for Qwen, where it contains the true pathology 12–24 points more often and gets about as close to the reference as five averaged base-model lists; severe conditions are named more often on average, but specific ones, most often acute pulmonary edema, are lost. Majority voting stops helping and agreement between samples weakens as an error signal; a label-free rank average of agreement and stated probability beats either alone. Qwen3-8B's thinking mode restores agreement as the better error signal and brings pass@32 back to the base model's level, though not the breadth of its answers. Studies of diagnostic breadth should report which readout they use and whether the model reasons first.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022b) (synthetic patients) and Phenopacket Store v0.1.27 (Danis et al., 2025) (cases from published literature). All models are public checkpoints (Table 4). Prompts and an example vignette are in Appendix A; code and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read

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in two ways: from the list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty (Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples (Yue et al., 2025; Wu et al., 2025; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Instruction-tuned models do not: they verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026; Jang et al., 2026), and one aligned answer can cover what many base-model samples spread out (Lake et al., 2025). We ask what this means for differential diagnosis, where every case has a reference differential and a true diagnosis, and what it does to the tools that read the differential from samples. This is a replication and extension in a clinical setting, not a new phenomenon.

The design holds the pretrained weights fixed and varies only the post-training: base/instruct pairs for Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; three medical models fine-tuned from Qwen2.5-7B-Instruct; and Qwen3-8B with its thinking mode on. Each final pair is also run with worked examples in the prompt, so that base models answer in the requested format. Our measurements:

- **Post-training keeps the base model's view and removes its variation.** A post-trained list tracks its own base model's averaged list for the same patient and barely changes between calls (§4.2).
- **Sampled answers collapse** in every family, by 13–30 points of pass@32 with each model at its best prompt; part of this is sharper sampling for one model, and the stage that narrows most depends on the recipe (§4.1).
- **The stated list improves robustly only for Qwen**, by naming the true pathology and the reference's conditions more often; severe conditions are named more often on average, but specific ones are lost (§4.2).
- **Sampling-based methods lose value**: voting stops helping and agreement weakens; a label-free rank average of the two signals is a sound default, and thinking mode restores agreement (§4.3). On real rare-disease cases the voting and agreement shifts replicate, but agreement stays the better signal (§4.4).

2. Related work

Post-training and the sampled distribution. RL with verifiable rewards improves pass@1 while base models win at large k on math and code (Yue et al., 2025; Wu et al., 2025); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh et al., 2026), though RL can expand the boundary in domains under-represented in pretraining (Cheng et al., 2025). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025; Murthy et al., 2025), more so with scale for SFT (Springer et al., 2026), and concentrate next-token distributions so that outputs barely

respond to decoding (Yang et al., 2026). Where diversity collapses depends on the pipeline: at DPO in OLMo and OLMo 2 (Peeperkorn et al., 2025) and in a stage ablation by Liu (2026), at each OLMo-2 stage for stated-versus-sampled distributions (Jang et al., 2026), and set by training data across Olmo 3's Think, Instruct and RL-Zero lineages (Karouzos et al., 2026). Base models given a few in-context examples approach aligned models (Lin et al., 2023).

Stated versus sampled distributions and uncertainty. Models asked to verbalize a distribution match it better than their samples do (Meister et al., 2025; Jang et al., 2026), and they struggle to draw from target distributions directly (Gu et al., 2026; Zhao et al., 2026). Verbalized sampling keeps most of the base model's diversity through Tülu 3's checkpoints (Zhang et al., 2026), and aligned answers aggregate content that base models spread across samples on open-ended questions with conflicting perspectives (Lake et al., 2025). Our stated readout is the verbalized top-k elicitation of Tian et al. (2023). For error detection, sample consistency usually beats verbalized confidence and hybrids beat both (Xiong et al., 2024; Lyu et al., 2024), including for diagnosis (Savage et al., 2025; Ren et al., 2026); aligned models' samples can collapse enough to leave sampling-based uncertainty without signal (Liu, 2026), and cross-model disagreement (Hamidieh et al., 2026) or the base model (Luo et al., 2025) can supply what self-consistency misses.

Medical models and differential diagnosis. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases (Kanjee et al., 2023; McDuff et al., 2025) and on Phenopacket Store, where they trail the phenotype-driven tool Exomiser (Reese et al., 2026; Chimirri et al., 2025). Medical post-training includes verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). DDXPlus scores predicted against reference differentials as sets (Fansi Tchango et al., 2022a); H-DDx proposes hierarchical F1 (Lim et al., 2025), MEDDxAgent studies interactive diagnosis (Rose et al., 2025), and Puri et al. (2026) show on DDXPlus with Qwen3-8B that single-answer RL collapses sampled answers and train models to output several answers per completion. Relative to Jang et al. (2026), who compare stated and sampled distributions on synthetic and survey targets with two of our pairs, Puri et al. (2026) and Karouzos et al. (2026), we add a reference distribution with a true diagnosis, a direct test that post-trained lists reproduce the base model's per-case view, prompt and temperature controls with condition-clustered statistics, and the consequences for voting, error signals and severe conditions.

3. Setup

Tasks. DDXPlus (Fansi Tchango et al., 2022b): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, the engine's posterior over 49 conditions. We use 980 test patients (20 per condition) rendered by template as questionnaire-style vignettes without exam or test results (example in Appendix A). The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON, so this is a closed-set task. **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases covering 749 diseases (at most two per disease), ren-

dered as phenotype vignettes with an open-ended differential; names are graded against the case's diagnosis via Mondo as exact, related (a specific ancestor group or subtype) or wrong (Appendix A).

Models and sampling. Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B-v0.3, each with its official post-trained version; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT, DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General, trained with RL from the 7B base; and AlphaMed, HuatuoGPT-o1 and m1, fine-tuned from Qwen2.5-7B-Instruct (Table 4). We draw 32 samples per case with vLLM at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (base models without one get a post-trained sibling's) and Qwen3's thinking mode off unless stated. A sample's *top answer* is valid if it is one of the 49 conditions and its *list* if it names at least one; invalid answers count as wrong. With this zero-shot prompt base models are often invalid (up to 47% of samples for OLMo and 76% for Mistral), so every final pair is re-run with three worked examples from the training split (Appendix D.3), on which every model is valid in over 99% of samples.

Readouts. The *sampled* readout is the distribution of top answers over a case's samples: pass@k (Chen et al., 2021), distinct top answers, the reference mass they cover (coverage@32), and total-variation distance (TV) of their empirical distribution to the reference. The *stated* readout is one sample's list, with its probabilities renormalized over the conditions it names (uniform if missing), averaged per case over valid samples: whether it contains the true pathology among its top k (recall@k), its TV to the reference (also with the true pathology removed from both), the reference mass it names, its *conditional* TV (to the reference restricted to the named conditions and renormalized, a scale-free measure of how it allocates probability), DDXPlus's set F1 against the reference's support (DDF1), and whether it names each severe condition (DDXPlus severity 1–2, the dataset's label) that the reference weights at least 0.1. The *single-call answer* is the top answer of a model's first valid sample.

Statistics. DDXPlus has 20 cases per true pathology, so intervals are 95% percentile intervals from a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples) unless stated; p-values use the add-one correction, and the headline base-vs-post differences are Holm-corrected over 112 tests, other families separately (Appendix G). Appendix intervals marked as case-level are two to three times narrower, and some appendix tables come from an earlier scorer whose values differ by up to 0.003. AUROCs use average ranks for ties. With five to six pipelines, results describe these checkpoints rather than post-training in general.

4. Results

4.1. Post-training collapses the sampled answers

Table 1 compares each base model with its post-trained versions. At 7–8B the post-trained Qwen model is far better on its first answer (+12.6 to +27.5 points), but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 9.7–14.6 points (Figure 2 in Appendix C.1). At 32B the zero-shot losses are smaller and uncertain (Qwen2.5-

32B -4.9 $[-10.2, +0.4]$, OLMo 3.1 32B -4.0 $[-6.7, -1.3]$, neither surviving correction). Base models answer the zero-shot prompt invalidly far more often, and some post-trained models do worse with worked examples (Qwen2.5-7B-Instruct 0.598 → 0.535), so we also compare each base model with worked examples against its post-trained model at its better prompt: post-training then costs 13.2–29.9 points of pass@32 in all six final pairs, including both 32B pairs and Mistral-7B, also with example lists taken from another model instead of the reference (Appendices D.3–D.5). Part of this is sharper sampling. Zero-shot, Qwen2.5-7B-Instruct at T=1.3 matches its base model's pass@32 at T=0.7 (0.704 against 0.695), though not at T=1 (0.766), at a cost of 4.2 points of pass@1, whereas Qwen3-8B stays below its base model even at T=1.6 (0.684 against 0.732; Appendix D.4). Post-trained models give 1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9. Coverage@32 rewards spreading answers (a uniform guesser covers 0.48, more than any model), but TV does not: samplers that always give each case's reference mode or its true pathology score 0.694 and 0.714, better than a guesser that ignores the patient (0.774; Table 2). Base models' samples lie at 0.73–0.76, and post-trained samples are farther than the one-answer oracles in every setting and farther than the patient-blind guesser for three of the five final models (0.78–0.85). Which stage narrows the samples most depends on the recipe: in OLMo 3 7B the RLVR step costs 19.1 $[13.8, 24.6]$ points of pass@32 against the DPO model, in OLMo 3.1 32B the SFT step costs 12.1 $[6.8, 18.0]$ and RLVR 2.8 $[0.5, 5.3]$, and RL-Zero, trained with RL from the 7B base, loses 13.3 $[7.3, 19.7]$ points at T=0.7 but none detectably at T=1 (Appendices C.4, C.5). Qwen3-8B with thinking on reaches its base model's pass@32 (0.815 against 0.732 zero-shot and 0.809 with worked examples) with samples as close to the reference as the base model's, though still narrow (1.76 distinct answers; Appendix D.6).

4.2. Post-training keeps the base model's view and removes its variation

What is a post-trained list? For each patient we compare each post-trained list with the mean of its base model's lists for the same patient (Table 19). In all 15 settings the post-trained list is closer to that average than to the base model's averages for other patients with the same true pathology (TV -0.028 to -0.076) and far closer to it than to the reference (by 0.14–0.33). It is about as close to the base average as a single base-model list at T=0.7 (Qwen3-8B 0.368 against 0.372) and closer at T=1, but post-trained lists barely vary between calls (TV 0.11–0.33 to the mean of the model's other lists, against 0.37–0.69 for base lists). Post-training thus keeps the base model's case-specific view and removes its variation between calls. OLMo 3 7B is the exception: its lists drift from its base model's view (0.628 against 0.493).

Whether this improves the list depends on the model (Table 2). For every Qwen model and OLMo 3.1 32B, the post-trained list contains the true pathology among its top five 12–24 points more often (e.g. Qwen3-8B 0.67 → 0.80), is closer to the reference (TV -0.043 to -0.088; -0.018 to -0.056 without the true pathology, Qwen3-8B at T=0.7 not surviving correction), names more reference mass, and allocates probability among the named conditions more like the reference (conditional TV -0.025 to -0.078). OLMo 3 7B's lists do not improve (recall@5 0.47 → 0.49), and without the true pathology they are farther ($+0.034$ at T=0.7). With worked examples the Qwen gains in recall and TV

Comparison	Model	Sampled top answers pass@1	pass@32	Distinct	Cov@32	Drawn from stated lists pass@32	Distinct
Qwen3-8B, T=0.7	base	0.388	**0.732**	**3.77**	**0.323**	0.780	7.79
	post-trained	**0.559**	0.607	1.27	0.217	**0.811**	4.96
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.832**	9.82
	post-trained	**0.556**	0.643	1.53	0.236	0.820	5.40
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	0.772	10.68
	instruct	**0.478**	0.598	1.82	0.239	**0.802**	6.81
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	0.785	12.44
	instruct	**0.464**	0.663	2.65	0.290	**0.827**	7.76
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	0.850	8.79
	instruct	**0.604**	0.746	1.68	0.276	**0.880**	6.49
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	**0.719**	8.66
	SFT+DPO+RLVR	0.265	0.422	2.37	0.195	0.577	5.65
	RL-Zero (from base)	**0.295**	0.572	4.06	0.286	0.701	9.66
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	0.839	8.11
	SFT+DPO+RLVR	**0.560**	0.756	2.15	0.283	**0.853**	6.75
Chance	uniform guesser	0.020	0.48	23.7	0.48		
	reference-prior guesser	—	0.44	21.9	0.56		

Table 1: Sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top answers per case; Cov@32: reference mass they cover. *Drawn from stated lists*: each sample's top answer replaced by a draw from its own stated list. Bold: better model of each pair (OLMo: of the block; the SFT and DPO stages are in Tables 8 and 9). Invalid answers count as wrong, which mostly penalizes base models. Chance: guessers that ignore the patient, drawing uniformly from the 49 conditions or from the reference differential averaged over test cases.

persist, OLMo 3.1 32B keeps only a higher reference mass, and Mistral's gain depends on the examples (TV −0.036 with the reference's lists, −0.059 with another model's); the stated-list gain is robust only for Qwen. Averaging calls closes most of the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, OLMo 3.1 32B 0.691 against 0.701; diamonds in Figure 1a). With five lists on both sides, post-training is closer in four of eleven direct-answer settings (all Qwen), not detectably different in four and farther in three (OLMo; Appendix D.7). Qwen3-8B with thinking on goes further: one of its lists (TV 0.670) beats five averaged base lists.

Severe conditions. Post-trained lists name the severe conditions that the reference weights at least 0.1 more often in every zero-shot setting (Qwen +9.0 to +13.8 points, OLMo 3 7B +6.0 and +8.7; OLMo 3.1 32B +2.2, n.s.), and the mean inclusion over all 17 severe conditions rises in seven of eight. Yet one list names only 0.32–0.47 of the warranted ones, against 0.975 for the reference's top five, and specific conditions are lost (Table 11): acute pulmonary edema is the largest loss in four of the five final pairs (e.g. Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31). With worked examples the mean inclusion falls for Qwen2.5-32B and OLMo 3 7B, and five averaged post-trained lists name fewer warranted severe conditions than five averaged base lists in six of eleven settings (by 0.06–0.14). The Qwen models shift toward acute coronary syndromes and pulmonary embolism, which emergency medicine rules out first, but these vignettes lack the ECG and biomarkers

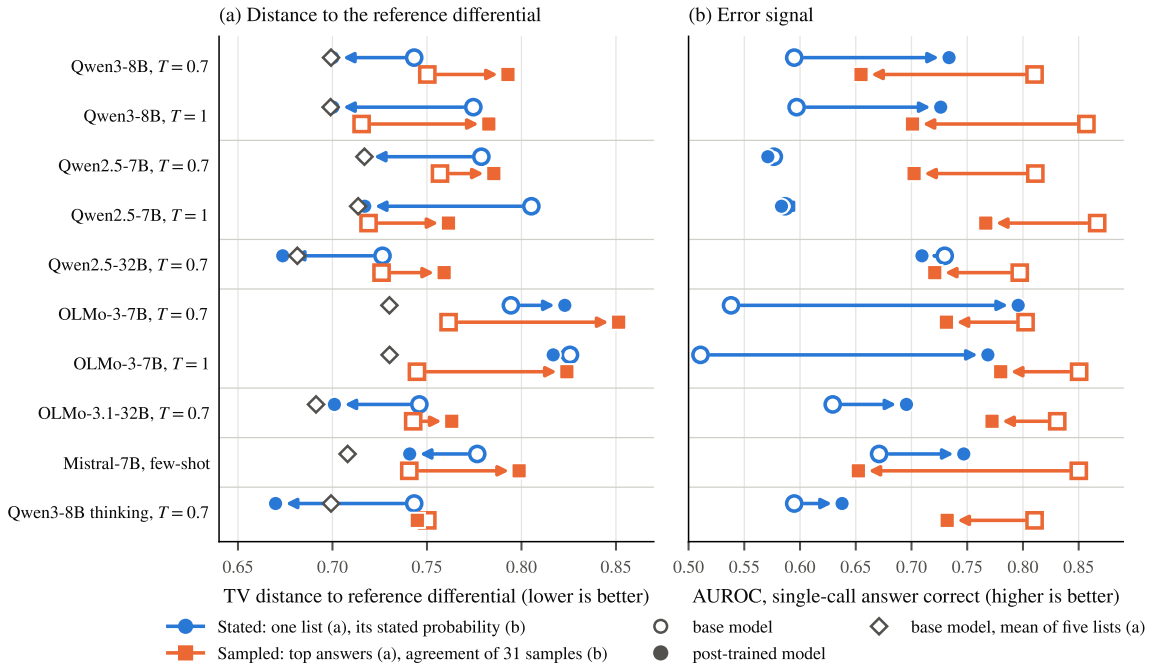


Figure 1: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; Mistral-7B with the few-shot prompt; the last row compares Qwen3-8B-Base with Qwen3-8B with thinking on). (a) TV to the reference differential of one stated list (blue circles) and of the sampled top answers (orange squares); grey diamonds: the mean of the base model's first five lists. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

that separate the coronary syndromes, so the shifts track the reference engine rather than clinical correctness.

4.3. Consequences for sampling-based methods

Aggregation. Majority voting over 32 samples (self-consistency) raises base models' accuracy by 3.4–8.4 points at T=0.7 but adds at most 1.9 for the final post-trained models (Qwen2.5-32B-Instruct, [0.8, 3.1]; the others n.s.; Table 3). Models that reason before answering still gain (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with a CoT prompt +4.5), and so does Qwen2.5-7B-Instruct sampled at T=1.3 (+5.6). For a top-three differential, the top three of one list contains the true pathology more often than the three most frequent of 32 samples for the post-trained Qwen models (+5.6 to +13.7 points), while for every base model the samples win by 7.3–13.6.

Uncertainty. We score each signal by how well it ranks the correctness of the single-call answer (AUROC); the stated probability needs no extra calls, *agreement* (the share of 31 further samples with the same top answer) needs them. For base models agreement beats

Comparison (T=0.7)	Model	TV to reference↓ One list	Samples	List w/o gold	Reference mass named	Cond. TV↓	DDF1	Recall @5	Severe named
Qwen3-8B	base	0.743	**0.750**	0.833	0.334	0.492	0.276	0.666	0.347
	post-trained	**0.700**	0.793	**0.815**	**0.396**	**0.431**	**0.318**	**0.802**	**0.437**
Qwen2.5-7B	base	0.779	**0.757**	0.856	0.291	0.553	0.245	0.573	0.345
	instruct	**0.717**	0.785	**0.816**	**0.364**	**0.474**	**0.307**	**0.735**	**0.439**
Qwen2.5-32B	base	0.726	**0.726**	0.830	0.351	0.473	0.282	0.721	0.382
	instruct	**0.674**	0.759	**0.787**	**0.418**	**0.437**	**0.333**	**0.840**	**0.472**
OLMo 3 7B	base	**0.794**	**0.761**	**0.850**	**0.267**	**0.568**	**0.225**	0.470	0.256
	Instruct (RLVR)	0.823	0.851	0.884	0.257	0.604	0.220	**0.485**	**0.317**
OLMo 3.1 32B	base	0.746	**0.743**	0.835	0.334	0.497	0.273	0.652	0.370
	Instruct (RLVR)	**0.701**	0.763	**0.812**	**0.391**	**0.467**	**0.312**	**0.779**	**0.392**
Chance	patient-blind	0.794	0.774	—	0.238	—	0.272	0.102	0.149
Oracle	reference	0.265	0.18–0.25	—	0.735	0.000	—	—	0.975
	one answer: reference mode	—	0.694	—	—	—	—	—	—
	one answer: true pathology	—	0.714	—	—	—	—	—	—

Table 2: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (per-case mean over valid samples). *Samples*: empirical distribution of top answers. *List w/o gold*: TV after removing the true pathology from list and reference. *Cond. TV*: TV between the list and the reference restricted to the named conditions and renormalized (how the list allocates probability, independent of how much mass it names). DDF1: set F1 of the named conditions against the reference's support. *Recall@5*: the list contains the true pathology. *Severe named*: share of severe conditions with reference probability ≥ 0.1 that one list names. Chance: the top five of the average reference differential as a fixed list, and 32 draws from that average as samples. Oracle: the reference's own top five as the list and as many reference draws as valid samples; and samplers that always give one answer, each case's reference mode or its true pathology. Bold: better model of each pair.

the stated probability by 0.07–0.26. Post-training weakens it in every family (Qwen3-8B 0.810 → 0.655, Qwen2.5-7B 0.811 → 0.702, Qwen2.5-32B 0.797 → 0.721, OLMo 3 7B 0.802 → 0.731, OLMo 3.1 32B 0.831 → 0.772; Figure 1b), significantly for Qwen but not OLMo, consistent with Liu (2026); since collapsed samples often tie, which pushes agreement's AUROC toward 0.5, this largely restates the collapse. The stated probability then beats agreement for Qwen3-8B (+0.079 [0.011, 0.143]), and with worked examples for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral-Instruct (+0.095 to +0.130), tying for the others: the better signal depends on model and prompt. Thinking returns Qwen3-8B to the base pattern (agreement better by 0.094 [0.027, 0.170]), and so does sampling Qwen2.5-7B-Instruct at T=1.3 (0.794 against 0.602).

Which signal to use. Across 56 model–setting combinations (Appendix D.8), averaging the two signals' ranks, with no labels, gives a held-out AUROC of 0.746 for the 39 post-trained combinations, above always using agreement (+0.051 [0.022, 0.074]; better in 29 of 39) or the stated probability (+0.075 [0.058, 0.098]; 39 of 39); a logistic combination fitted on 100 labeled cases adds only 0.007 [0.001, 0.015] (intervals resample the 26 underlying models). Neither signal tracks how ambiguous a case is (Spearman with the reference's entropy between −0.11 and +0.16; Table 20).

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen3-8B (thinking)	0.672	+1.5	+3.0	0.638	0.693	**0.732**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 3: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call accuracy uses the first valid sample, so it can exceed pass@1, which counts invalid answers as wrong. Vote@32 gain: accuracy of the majority top answer over all valid samples minus single-call accuracy (points). Top-3: rate at which one list's top three contains the true pathology minus the rate for the three most frequent sampled answers (points). AUROC: stated probability of the single-call answer (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal.

4.4. Real rare-disease cases and medical post-training

Real rare-disease cases. Phenopacket Store has no reference differential, and 7–8B models are near the floor (pass@32 2.5–7.0%, against about 20% top-1 for GPT-4o (Chimirri et al., 2025)), so we count an answer as correct if it is the diagnosis or a closely related Mondo class (36–130 correct single-call answers per model; Appendix E.3) and read these results as directional. The voting and agreement shifts replicate: majority voting helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5, and post-training weakens agreement (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Unlike on DDXPlus, agreement stays the better signal and the three most frequent sampled answers still beat one list's top three (Qwen3-8B: a tie).

Medical models. Medical post-training on Qwen2.5-7B-Instruct does not narrow the samples further: AlphaMed improves on its parent at large k (+8.4 [2.6, 14.7] points at k=32) with unchanged diversity, and HuatuoGPT-o1 and m1, trained on reasoning traces, widen the samples (2.49 and 3.29 distinct answers, against 1.82; Appendix E.1). On Phenopacket Store the RL-trained medical models raise stated confidence from 0.51 to 0.67–0.69 while accuracy gains stay under 1.5 points (Appendix E.2).

5. Discussion and limitations

Jang et al. (2026) show that instruction-tuned models cannot sample from distributions they can describe. In differential diagnosis, against a reference with a known true pathology, we find the same split and a simpler account of it: direct-answer post-training keeps the base model's view of each patient and removes most of its variation between calls. The sampled answers therefore collapse, and evaluations that read the differential from samples (pass@k at large k, diversity, self-consistency, agreement-based uncertainty) report a loss that a clinician reading one answer would not see; a label-free rank average of agreement and stated probability is a reasonable default error signal. For Qwen, one post-trained call is better than one base call, naming the true pathology and the reference's conditions more often, but no better than averaging a few base calls, so it is no evidence that post-training adds diagnostic knowledge. Every model stays far from the reference, neither readout tracks how ambiguous a case is, and losses of specific severe conditions, notably acute pulmonary edema, need checking model by model. These effects belong to answering directly at the usual temperatures: thinking mode restores pass@32 and agreement as an error signal, though not breadth, and hotter sampling does much the same for Qwen2.5-7B-Instruct but not Qwen3-8B. Which stage narrows the samples most is a property of the recipe.

Limitations. DDXPlus is synthetic and closed-set: its reference differential is a rule-based engine's posterior over questionnaire answers without exam or test results, sometimes clinically implausible (Appendix A), its test set is class-balanced, and the prompt lists all 49 conditions, so our metrics measure agreement with that engine, not clinical correctness, and no clinician reviewed lists or references. Severity is the dataset's label, and the prompt asks for the most likely conditions, not must-not-miss ones. The few-shot controls use one temperature and one set of example patients; the temperature control is zero-shot and covers two 7–8B models. We study open 7–32B models from three families, one reasoning mode and no frontier models, and the 32B models only at T=0.7. Phenopacket accuracy is near the floor, and its grader is rule-based and not validated against clinicians.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

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Appendix A. Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster

headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12; cluster headache is implausible for this patient, a reminder that the reference is an engine's posterior rather than a clinician's differential):

```
A 13-year-old female presents for evaluation. Symptoms (patient
answers): Do you have pain somewhere, related to your reason for
consulting? Yes. Characterize your pain: burning. Do you feel pain
somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is
the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes.
Do you have nasal congestion or a clear runny nose? Yes. Do you have a
cough? Yes. Relevant history and risk factors: Have you had a cold in
the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]
```

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§D.1) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §C.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 4: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Appendix B. Models

Appendix C. The sampled differential in detail

C.1. General post-training trades coverage for top-1 accuracy

Table 5 and Figure 2 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.3, +12.0]$.

C.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.7 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 5). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 5: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

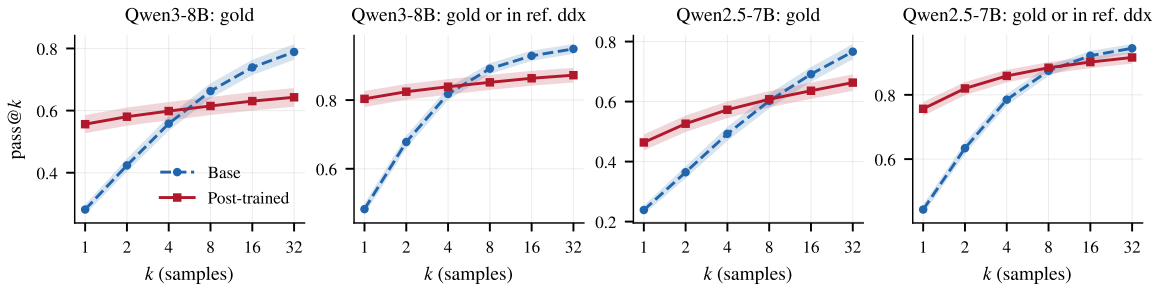


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). This concerns verbalized confidence; alignment is known to worsen the calibration of token probabilities (OpenAI, 2023; Zhu et al., 2023), while verbalized confidence of RLHF models is better calibrated (Tian et al., 2023). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 6), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one

that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess severe conditions on the stated list instead (§D.2). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives. Ruling out acute coronary syndrome first is standard practice for chest pain, but these vignettes lack the ECG and biomarkers that would separate the coronary conditions, so we read this as a concentration of the samples, not as a clinical error.

C.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 7). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains $+23.0$ $[+20.9, +25.1]$ points at k=1 and loses -7.1 $[-9.3, -5.0]$ at k=32 (T=0.7; at T=1, $+34.2$ and -8.7 $[-11.0, -6.4]$), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 ($+4.7$ $[+2.9, +6.6]$ at k=32) and ties it at T=1 (0.0 $[-2.1, +1.9]$). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§E.1).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain $+12.0$ $[+9.8, +14.1]$ (Qwen3, T=0.7), $+18.7$ $[+16.6, +20.7]$ (Qwen3, T=1), $+12.2$ $[+10.4, +14.0]$ and $+16.8$ $[+15.0, +18.5]$ points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid,

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 6: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the severity-1–2 tier; medical post-training raises it; our GRPO from the base model loses at most 4.3 points, and with the group reward the severe-tier loss is not significant.

the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

C.4. Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 8 gives the results.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by −14.6 [−17.5, −11.7] points after SFT, −9.2 [−12.1, −6.1] after DPO, −28.3 [−31.5, −25.0] after RLVR, and −13.3 [−16.1, −10.5] for RL-Zero. Diversity falls at

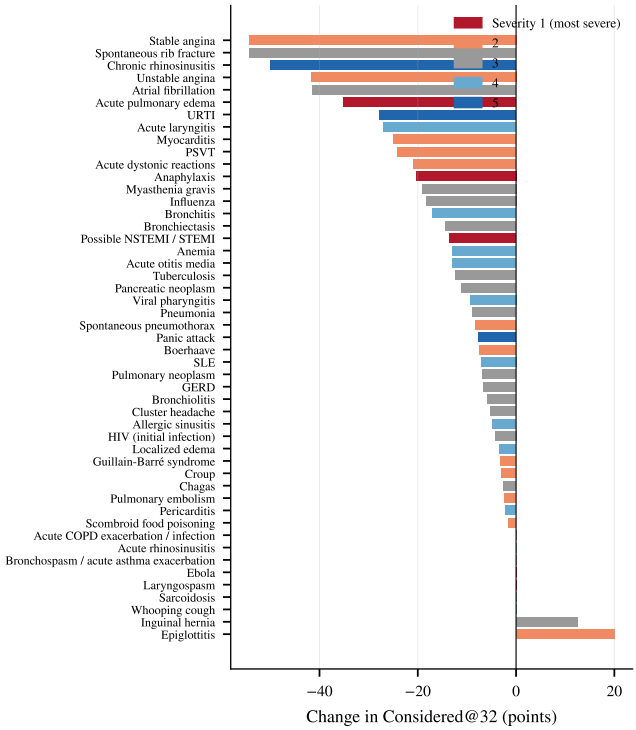


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k (+5.4 $[+2.9, +8.0]$ points at k=32, and +10.3 $[+9.0, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.8, -16.3]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-7.2, -4.6]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base, against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-15.4, -9.9]$ at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (-1.8 $[-4.5, +0.8]$; clustered $[-5.8, +2.1]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 (clustered $[7.3, 19.7]$), about as much as SFT.

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.558**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 7: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §C.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§C.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero at T=1, AlphaMed's base-initialized checkpoint, and our GRPO runs in §F) narrows the differential less, but not always: RL-Zero at T=0.7 loses about as much as SFT. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 $[-3.0, -1.1]$ points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 $[-1.4, +0.2]$), from a base model at 4.5% (§E.2).

C.5. Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 9). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[2.8, 7.1]$ points at pass@32 for Qwen2.5-32B and 4.0 $[2.0, 5.9]$ for OLMo 3.1 32B (clustered $[-0.6, 10.2]$ and $[1.2, 6.8]$; neither survives correction), halves the number of distinct answers ($3.78 \to 1.68$ and $4.32 \to 2.15$), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§D.3). The stage attribution differs from 7B. In OLMo 3.1 32B

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.266	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 8: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 6. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 9: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family.

the largest narrowing is SFT (pass@32 −12.1 [−14.4, −9.9], clustered [−18.0, −6.8], distinct answers −1.39, severe tier −11.9); DPO recovers most of it (+10.9 against SFT), and the final RL step is small (−2.8 [−4.3, −1.1] at k=32, clustered [−5.3, −0.5], −0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.2; with few-shot prompts only Qwen2.5-32B's does robustly, §D.3), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

Model	TV to reference↓ One distribution	TV to reference↓ Samples	AUROC, single-call answer correct Stated p	AUROC, single-call answer correct Agree@5	AUROC, single-call answer correct Agree@32	Top-3: one dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 10: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

Appendix D. The stated readout: elicitation, robustness and audit

D.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 10). The results match the short list. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038). Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

D.2. Ruling out format, accuracy and floor effects

Table 12 tests simpler explanations of the stated-list result. *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV across both temperatures; Table 12 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (−0.018 to −0.056), and DDXPlus's set

F1 against the reference's support improves by 0.040–0.083 (Table 24). TV, mass and DDF1 are not independent, since a list's TV is the reference mass it leaves out plus the mass it misallocates among the conditions it names, and the second term grows mechanically with the first. The conditional TV, between the list and the reference restricted to the named conditions and renormalized, removes this: it falls after post-training for every Qwen pair and OLMo 3.1 32B (−0.025 to −0.078, all surviving correction) and rises for OLMo 3 7B at T=1 (+0.064), so the Qwen lists both name better conditions and weight them more like the reference. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.74 → 0.65), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 11 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. That count test is weak against an overall upward shift, so Table 11 also gives the mean shift across the 17 conditions: it rises in seven of eight zero-shot settings but falls with worked examples for Qwen2.5-32B (−0.012) and OLMo 3 7B (−0.018), and is flat for Mistral (−0.009, p=0.05). Significant losses include acute pulmonary edema (Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model. Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.811 vs. 0.780 at T=0.7, Qwen2.5-32B-Instruct 0.880 vs. 0.850, OLMo 3.1 32B Instruct 0.853 vs. 0.839), but not for Qwen3-8B at T=1 (0.820 vs. 0.832) or OLMo 3 7B (0.577 vs. 0.719), and a uniform guesser reaches 0.48. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

D.3. Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different

Pair (base → post)	Named less / more often	Significant less / more	P(as many less often)	Mean shift (17 conditions)	P	Largest loss (shift)
Qwen3-8B, T=0.7	5 / 12	2 / 3	0.98	+0.040	0.001	acute pulmonary edema (-0.14)
Qwen3-8B, T=1	6 / 11	2 / 4	0.93	+0.060	0.001	stable angina (-0.06)
Qwen2.5-7B, T=0.7	5 / 12	2 / 3	0.98	+0.029	0.001	stable angina (-0.16)
Qwen2.5-7B, T=1	4 / 13	2 / 5	0.99	+0.065	0.001	stable angina (-0.13)
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	0.93	+0.047	0.001	acute pulmonary edema (-0.19)
OLMo 3 7B, T=0.7	8 / 9	7 / 6	0.71	+0.046	0.001	acute pulmonary edema (-0.24)
OLMo 3 7B, T=1	8 / 9	6 / 6	0.69	+0.061	0.001	NSTEMI/STEMI (-0.12)
OLMo 3.1 32B, T=0.7	7 / 10	3 / 0	0.82	+0.001	0.854	acute pulmonary edema (-0.16)
Qwen3-8B, few-shot	5 / 12	1 / 4	0.97	+0.061	0.001	acute pulmonary edema (-0.16)
Qwen2.5-7B, few-shot	8 / 9	4 / 2	0.69	+0.034	0.001	stable angina (-0.15)
Qwen2.5-32B, few-shot	10 / 7	5 / 3	0.29	−0.012	0.009	laryngospasm (-0.25)
OLMo 3 7B, few-shot	9 / 8	7 / 3	0.51	−0.018	0.018	croup (-0.51)
OLMo 3.1 32B, few-shot	9 / 8	4 / 2	0.50	+0.011	0.012	stable angina (-0.25)
Mistral-7B, few-shot	11 / 6	3 / 0	0.17	−0.009	0.052	pulmonary embolism (-0.31)

Table 11: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; post-trained minus base, with intervals that resample the true pathologies (10,000 resamples) and Benjamini–Hochberg correction across the 17 conditions. P columns compare the count of conditions named less often and the mean shift across conditions with a null that swaps each case's base and post-trained lists at random (2,000 swaps); the count test is weak against an overall upward shift, so the mean shift and the largest single loss are shown too.

true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 13).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 14): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV −0.057 to −0.087, −0.055 to −0.073 with the true pathology removed, and DDF1 +0.051 to +0.081, all surviving correction. OLMo 3 7B remains the exception (TV +0.021, n.s.; +0.024 without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names (+0.047), but its TV (−0.022), TV without the true pathology (−0.006) and DDF1 (+0.009) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1	pass@32 list-sampled
Qwen3-8B-Base	0.90	0.743	0.663	0.806	0.479	**0.557**	0.752	0.780
Qwen3-8B	0.99	**0.701**	**0.646**	**0.769**	**0.435**	0.613	**0.680**	**0.811**
Qwen2.5-7B	0.95	0.779	0.689	0.828	0.514	**0.575**	0.823	0.772
Qwen2.5-7B-Instruct	0.91	**0.719**	**0.655**	**0.773**	**0.452**	0.605	**0.796**	**0.802**
Qwen2.5-32B	0.96	0.725	0.661	0.795	0.462	**0.543**	0.735	0.850
Qwen2.5-32B-Instruct	0.98	**0.674**	**0.639**	**0.726**	**0.409**	0.579	**0.645**	**0.880**
OLMo-3-1025-7B	0.81	**0.794**	0.687	**0.837**	**0.524**	**0.522**	**0.747**	**0.719**
OLMo-3-7B-Instruct	0.95	0.823	**0.659**	0.882	0.558	0.671	0.779	0.577
OLMo-3-1125-32B	0.89	0.744	0.662	0.812	0.481	**0.532**	0.706	0.839
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.644**	**0.774**	**0.436**	0.583	**0.634**	**0.853**

Table 12: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. A uniform guesser scores about 0.48 at pass@32.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68); against agreement across 31 samples the stated probability wins clearly only for Qwen3-8B and Qwen2.5-7B-Instruct (+0.108 and +0.130, clustered) and ties for the others. This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

D.4. Two further controls

Reference-free few-shot examples. The examples of §D.3 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 15). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and, now clearly, for Mistral (TV −0.053 to −0.077; −0.031 to −0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1 32B, and wherever the list improves its allocation among the named conditions also moves closer to the reference (conditional TV −0.051 to −0.100). Voting helps base models by 3.7–13.7 points and post-

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5	AUROC agree@32
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828	0.862
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563	0.584
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792	0.827
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594	0.626
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838	0.875
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647	0.694
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823	0.871
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668	0.696
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826	0.865
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684	0.744

Table 13: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §D.3). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 or the next 31 samples. Bold: better of the pair.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 14: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix G).

trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more severe conditions less often than chance. Base models score lower at k=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

Matched temperature. Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at T=1.3 and 1.6 (Table 16), and the two models differ. Qwen3-8B barely broadens: at T=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at T=0.7 (0.732) and at T=1 (0.789; −10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at T=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at T=0.7 (0.704 against 0.695), gains 5.6 points from

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Cond. TV
Qwen3-8B	−14.6	+0.049	−0.077	−0.049	+0.079	−0.078
Qwen2.5-7B	−19.7	+0.031†	−0.077	−0.059	+0.074	−0.094
Qwen2.5-32B	−14.5	+0.036	−0.053	−0.057	+0.054	−0.051
OLMo 3 7B	−29.7	+0.116	−0.007†	+0.013†	−0.002†	−0.015†
OLMo 3.1 32B	−17.4	+0.073	−0.016†	+0.002†	+0.009†	−0.005†
Mistral-7B	−12.7	+0.038†	−0.059	−0.031	+0.057	−0.100

Table 15: Few-shot control with reference-free examples (DDXPlus, T=0.7, 980 cases): the same three training patients as in §D.3, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; conditional TV Holm-corrected as a separate family).

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV samples	TV one list	Vote@32 gain	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.7	11.3	0.388	0.732	3.77	0.750	0.743	+3.4	0.595	0.810
	1	22.3	0.282	0.789	6.60	0.715	0.775	+10.4	0.597	0.857
Qwen3-8B	0.7	0.0	0.559	0.607	1.27	0.793	0.700	+0.3	0.734	0.655
	1	0.0	0.556	0.643	1.53	0.783	0.700	+1.4	0.726	0.701
	1.3	0.0	0.552	0.668	1.73	0.776	0.701	+1.0	0.707	0.724
	1.6	0.0	0.549	0.684	1.92	0.770	0.701	+1.8	0.715	0.745
Qwen2.5-7B	0.7	0.9	0.352	0.695	4.91	0.757	0.779	+5.2	0.576	0.811
	1	18.4	0.239	0.766	8.23	0.719	0.805	+14.1	0.587	0.867
Qwen2.5-7B-Instruct	0.7	0.0	0.478	0.598	1.82	0.785	0.717	+0.8	0.571	0.702
	1	0.0	0.464	0.663	2.65	0.761	0.717	+4.4	0.583	0.767
	1.3	2.2	0.436	0.704	3.38	0.745	0.722	+5.6	0.602	0.794
	1.6	30.5	0.304	0.726	3.88	0.729	0.749	+10.0	0.637	0.814

Table 16: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-p 1) against their base models at T=0.7 and T=1. Columns as in Tables 1, 2 and 3.

voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at T=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

D.5. A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. (2026). With the zero-shot prompt the base model does not answer: 52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §D.3 the base

model answers validly (0.4% invalid), and the pair reproduces the other families (Table 17). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model (0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest significant gain of the three families, and 11 of the 17 severe conditions are named less often. Mistral-Instruct also names 0.4 conditions per list that are not among the 49.

D.6. Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about 1,500 tokens and all end before the limit), the same weights behave differently (Table 17). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above the non-thinking model (0.607) and level with the base model (0.732 zero-shot, +8.4 $[1.7, 15.3]$ points, not surviving Holm correction over 42 tests; 0.809 with worked examples). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement across 31 samples is again the better signal (AUROC 0.732 against 0.638; difference 0.094 $[0.027, 0.170]$), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §C.3.

D.7. Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 18). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–28.8 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). One post-trained call thus gets about as close to the reference as several base calls; the packing test (Table 19) shows that it is a stabilized version of

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	TV, one list DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 17: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §D.3. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 13; Distinct: distinct top answers per 32 samples. Bold: best within each block.

the base model's view of the patient rather than its literal average, and, as for the open-ended questions of Lake et al. (2025), post-training adds no information that averaged base lists lack. Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming one to three more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.48), so the coverage deficit of §4.1 is one of the sampled readout.

D.8. Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. Combining the signals beats choosing between them. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39 post-trained combinations.

Pair	TV, base model 1 list	TV, base model mean of 5	TV, base model mean of 32	TV, post-trained model 1 list	TV, post-trained model mean of 5	TV, post-trained model mean of 32	Post − base, 5 lists each TV	Post − base, 5 lists each Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	$-0.004^{\dagger}$	$+0.030^{\dagger}$
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	$+0.023^{\dagger}$
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	$-0.037^{\dagger}$
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	$-0.014^{\dagger}$	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	$-0.006^{\dagger}$
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	$+0.012^{\dagger}$
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	$-0.015^{\dagger}$	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	$+0.015^{\dagger}$	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 18: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 2) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

This matches the general finding that hybrids of consistency and verbalized confidence beat either (Xiong et al., 2024). For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

Appendix E. Medical post-training and rare disease

E.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]; clustered [−0.4, +8.4]), widening to +8.4 at k=32 ([+6.0, +10.6]; clustered [+2.6, +14.7]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24)

Pair	TV of a post-trained list to own base mean	TV of a post-trained list to same-diagnosis means	reference	Within-model benchmark base list	Within-model benchmark post list	Differences own − same dx	Differences own − base list
Zero-shot prompt							
Qwen3-8B, T=0.7	0.368	0.444	0.701	0.372	0.109	−0.076	−0.004†
Qwen3-8B, T=1	0.424	0.486	0.701	0.551	0.153	−0.062	−0.128
Qwen2.5-7B, T=0.7	0.464	0.519	0.717	0.470	0.248	−0.055	−0.006†
Qwen2.5-7B, T=1	0.532	0.575	0.717	0.650	0.315	−0.043	−0.119
Qwen2.5-32B, T=0.7	0.360	0.431	0.674	0.375	0.189	−0.071	−0.015†
OLMo 3 7B, T=0.7	0.628	0.658	0.823	0.493	0.238	−0.031	+0.135
OLMo 3 7B, T=1	0.651	0.679	0.818	0.691	0.326	−0.028	−0.040
OLMo 3.1 32B, T=0.7	0.386	0.452	0.700	0.432	0.227	−0.066	−0.046
Few-shot prompt							
Qwen3-8B	0.469	0.518	0.681	0.471	0.102	−0.049	−0.003†
Qwen2.5-7B	0.559	0.596	0.702	0.515	0.171	−0.037	+0.044
Qwen2.5-32B	0.465	0.511	0.654	0.441	0.154	−0.045	+0.024
OLMo 3 7B	0.669	0.710	0.808	0.594	0.259	−0.040	+0.075
OLMo 3.1 32B	0.482	0.532	0.697	0.469	0.150	−0.050	+0.014†
Mistral-7B	0.495	0.546	0.740	0.495	0.181	−0.051	−0.000†
Thinking on							
Qwen3-8B, thinking	0.412	0.468	0.669	0.372	0.189	−0.057	+0.040

Table 19: Does a post-trained list reproduce its base model's view of the patient? (DDXPlus, 980 cases.) TV between a post-trained list and the mean of the base model's valid lists for the same case (own base mean), the base model's means for the other cases with the same true pathology, and the reference. Within-model benchmarks: a base list against the mean of the base model's other lists for the case, and a post-trained list against the mean of the post-trained model's other lists. †: the condition-clustered 95% interval includes zero.

Model (T=0.7)	Stated p	Agree@5	Agree@32
Qwen3-8B-Base	+0.042	+0.005	+0.024
Qwen3-8B	+0.116	+0.056	+0.098
Qwen2.5-7B	+0.044	+0.025	+0.038
Qwen2.5-7B-Instruct	+0.070	+0.054	+0.070
Qwen2.5-32B	+0.064	−0.026	−0.038
Qwen2.5-32B-Instruct	−0.043	−0.060	−0.109
OLMo-3-1025-7B	+0.083	−0.048	−0.052
OLMo-3-7B-Instruct	−0.067	−0.073	−0.049
OLMo-3-1125-32B	−0.005	+0.020	+0.011
OLMo-3.1-32B-Instruct	+0.063	+0.014	−0.023

Table 20: Spearman correlation between each uncertainty signal (lower confidence or agreement) and the entropy of the case's reference differential (DDXPlus, 980 cases). Neither readout tracks how ambiguous a case is.

stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 6). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 21: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model.

its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 21). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 $[+11.3, +16.0]$ and 18.1 $[+15.5, +20.7]$ points (clustered $[8.0, 20.0]$ and $[10.5, 25.9]$). m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 21, lower block). It does not. RL from the base model improves on the base at every k (+14.3 $[+13.1, +15.7]$ points at k=1, +9.5 $[+7.4, +11.6]$ at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 $[0.4, 3.9]$ points worse at k=1 and 10.8 $[+8.6, +13.2]$ points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (-0.7 $[-3.0, +1.7]$ points), against -12.4 for instruction tuning and -8.1 for instruction tuning followed by AlphaMed's RL (Table 6; all well below the chance levels of §C.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1 and in our GRPO runs (§F), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§C.4).

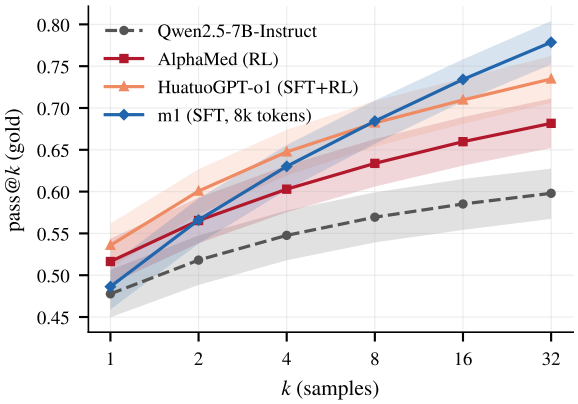


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

E.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every k ($+0.6$ $[+0.2, +0.9]$ points at k=1, $+1.0$ $[+0.1, +2.1]$ at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ ($+1.4$ $[+0.4, +2.5]$ at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 ($+0.7$ $[+0.4, +1.1]$) and trails it from k=8 on, by -2.2 $[-3.1, -1.3]$ points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we

Model	≤2022 (n=901) pass@1	≤2022 (n=901) pass@32	2023–24 (n=348) pass@1	2023–24 (n=348) pass@32	2025+ (n=152) pass@1	2025+ (n=152) pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 22: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§F) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 22). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

E.3. Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.3, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 23). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (-0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B ($+0.4$ $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct			Correct answers
				Stated p	Agree@5	Agree@32	
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 23: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 3; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

Appendix F. GRPO with a group-level reward

We also fine-tuned Qwen3-1.7B (instruct) and Qwen3-8B-Base with GRPO (LoRA, 200 steps, 16 samples per prompt, KL coefficient 0.04) on DDXPlus training patients disjoint from the test split, comparing the standard correctness reward (`acc`) with a group-level reward (`gm`). If k_a of the n samples in a GRPO group give top answer a, `gm` adds $\mathrm{clip}(n\,p_{\mathrm{ref}}(a) - (k_a - 1), 0, 1)$, which pays an answer only while its condition is under-represented relative to its reference mass. At 1.7B, a boosted variant (weight 3, training temperature 1.2, so the two changes are confounded) raised pass@1 by 1.6 and distinct answers by 0.10 over `acc` (two seeds; intervals ignore seed variation). From Qwen3-8B-Base, both recipes reached nearly the official Qwen3-8B's top-1 accuracy (0.537 and 0.539 vs. 0.558) while keeping more than twice its distinct answers (2.7–2.9 vs. 1.27), and `gm` narrowed the base model less than `acc` (distinct answers +0.14 at equal pass@1). Neither changed Phenopacket accuracy. Because `gm` needs a reference differential at training time, which real clinical data rarely provide, and the runs train on the evaluation distribution, these are existence results only. Closely related rewards computed from group answer counts have appeared since: a witness advantage that trains models to sample the distributions they state (Arabi et al., 2026), and a batch bonus that penalizes repeated answers (Song et al., 2025); we do not compare against them. The GRPO runs enter only the 56-combination audit (§D.8).

Appendix G. Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 24 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples).

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4	−2.50	+0.043	−0.043	−0.018†	+0.042	−0.160
	[−19.5, −5.9]	[−2.88, −2.13]	[+0.021, +0.063]	[−0.059, −0.028]	[−0.032, −0.005]	[+0.029, +0.056]	[−0.216, −0.097]
Qwen3-8B, T=1	−14.6	−5.06	+0.067	−0.074	−0.039	+0.066	−0.158
	[−21.6, −8.4]	[−5.49, −4.66]	[+0.042, +0.091]	[−0.090, −0.059]	[−0.052, −0.027]	[+0.053, +0.080]	[−0.211, −0.103]
Qwen2.5-7B,	−9.7	−3.09	+0.028†	−0.062	−0.041	+0.062	−0.106
T=0.7	[−16.2, −4.1]	[−3.61, −2.58]	[+0.007, +0.049]	[−0.076, −0.049]	[−0.054, −0.028]	[+0.049, +0.076]	[−0.162, −0.041]
Qwen2.5-7B, T=1	−10.3	−5.59	+0.042†	−0.088	−0.056	+0.083	−0.104
	[−17.4, −4.3]	[−6.04, −5.15]	[+0.015, +0.068]	[−0.103, −0.074]	[−0.069, −0.044]	[+0.069, +0.097]	[−0.160, −0.046]
Qwen2.5-32B,	−4.9†	−2.10	+0.033	−0.053	−0.043	+0.051	−0.079
T=0.7	[−10.2, +0.6]	[−2.50, −1.74]	[+0.018, +0.049]	[−0.064, −0.042]	[−0.053, −0.033]	[+0.041, +0.061]	[−0.119, −0.039]
OLMo 3 7B,	−28.3	−2.56	+0.090	+0.029†	+0.034	−0.006†	−0.076†
T=0.7	[−35.7, −21.2]	[−2.84, −2.27]	[+0.072, +0.109]	[+0.004, +0.052]	[+0.017, +0.051]	[−0.022, +0.009]	[−0.156, +0.000]
OLMo 3 7B, T=1	−15.2	−3.95	+0.079	−0.009†	+0.006†	+0.024	−0.076†
	[−20.9, −9.6]	[−4.22, −3.68]	[+0.060, +0.098]	[−0.030, +0.011]	[−0.006, +0.019]	[+0.010, +0.036]	[−0.150, −0.009]
OLMo 3.1 32B,	−4.0†	−2.18	+0.020†	−0.045	−0.023	+0.040	−0.056†
T=0.7	[−6.8, −1.2]	[−2.46, −1.89]	[+0.002, +0.037]	[−0.059, −0.031]	[−0.034, −0.012]	[+0.027, +0.053]	[−0.095, −0.018]

Table 24: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (10,000 resamples, add-one p-values). pass@32 here is recomputed from the samples with the analysis scripts' name matching and can differ from Table 5 by up to 0.3 points.

Two-sided p-values use the add-one correction, $2(1+\min(n_{\leq 0}, n_{\geq 0}))/(B+1)$, so the smallest attainable value, 2/10,001, lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs × 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests, marked †: the 32B pass@32 losses, the sampled-TV increases for Qwen2.5-7B and OLMo 3.1 32B, OLMo 3 7B's list TV at both temperatures and its list TV without the true pathology at T=1, Qwen3-8B's list TV without the true pathology at T=0.7, OLMo 3 7B's DDF1 at T=0.7, and the drop in agreement AUROC for both OLMo models. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.